Mustafa Kerem Kurban

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AI / MLOps Engineer | Cloud AI | Agentic Systems

AI/MLOps Engineer with 6+ years building production AI systems across finance, research, and applied ML. Deep hands-on experience across end to end agentic pipelines. Certified in LangChain, MCP / A2A, LLM fine-tuning (LoRA, Federated & Reinforcement Learning), governance & observability (Databricks, Unity Catalog, MLflow 3). Co-author on PLOS Biology 2024 and Network Neuroscience 2024. Holding Swiss B permit, no sponsorship required in Switzerland.

WORK EXPERIENCE

Wipro (Client: Large Swiss Bank), Zurich, Switzerland

Senior AI Engineer (Contract) *11/2025 – 06/2026*

- Embedded in Core Engineering Data & AI team, with deep collaboration with Risk and other AI teams to evaluate and integrate enterprise GenAI/MLOps capabilities, LLM evaluation frameworks (Databricks MLflow), CI/CD templates, model registries, and MCP-based data and tool access.
- Designed and shipped an agentic ingestion pipeline that standardized 300+ SOPs from reconciliation teams across global hubs, surfacing automation candidates and reducing manual SOP review effort by 90% with React based UI.
- Built an agentic benchmark on historical reconciliation breaks to measure agentic-LLM performance vs. ground-truth human resolutions, using MCP servers, SOP-derived knowledge bases backed with enterprise grade governance and observability best practices.
- Acted as cross-team enablement lead: ran enterprise AI-capability deep-dives for business and technical stakeholders across business and AI teams, cross-pollinated agentic and cloud patterns with internal and external practitioners.
- Standardised unstructured SOPs into governed ontologies for agentic and classical ML consumption, thereby grounding them with explainable and auditable agentic workflows.

Neptune.ai, Remote

AI researcher *03/2025 – 10/2025*

- Implemented post-training (SFT) and RLHF (DPO, PPO, GRPO) with prompt optimization (DSPy) on Mistral, Qwen3, and Llama 3.1 8B causing 210% improvement over base-model baseline on private-domain alignment tasks.
- Trained and benchmarked PEFT methods (rank-stabilized LoRA, Q-LoRA, DoRA) across dataset sizes, producing high-performance custom models with minimal compute overhead.
- Productionized LLMs via Ollama, Nvidia NIM, Azure AI Foundry, and AWS Bedrock — on-prem and cloud — using IaC (Terraform).
- Built A2A and MCP servers with hybrid RAG to orchestrate multi-agent platforms (LangChain, CrewAI, SmolAgents).

EPFL Blue Brain Project, Geneva, Switzerland

Machine Learning Engineer *04/2024 – 01/2025*

- Engineered a modular Python toolkit (Pydantic, LangChain, async APIs) for cross-modal data extraction, knowledge-graph search, and scalable LLM evaluation — reducing manual research effort by ~90%.
- Designed large-scale RAG pipelines (BM25, reranking, graph retrieval) for explainable LLM workflows, measurably reducing retrieval hallucinations in scientific Q&A and increasing user trust.
- Built LLM evaluation pipeline (RAGAS metrics + custom tool-calling benchmarks) feeding downstream RLHF workflows.
- Created a Neo4j knowledge graph from BBP citation networks (authors, articles, institutions, topics) combined with Microsoft GraphRAG to deliver an SME-facing chatbot.

Data Scientist *03/2020 – 04/2024*

- Engineered high-performance graph algorithms for triplet-motif detection in directed weighted neural networks, achieving 3 orders of magnitude speedup (hours ⟶ seconds).
- Deployed Dask-parallelized network statistics on large-scale graphs, enabling the world's largest hippocampus model (hippocampushub.eu), validated in PLOS Biology 2024 (doi:10.1371/journal.pbio.3002861).

Intel, Istanbul, Turkey

Machine Learning Ambassador *05/2017 – 09/2018*

- Co-published neuroimaging research with Stanford classifying anesthesia-exposed children from fMRI time-series at 80% accuracy using feature engineering and CNNs.
- Promoted Intel Movidius Neural Compute Stick VPU for edge inference in computer-vision pipelines; deployed and demonstrated edge models.

EDUCATION

MSc, Materials Science and Nanotechnology — Bilkent University, Turkey / EPFL, Switzerland

- Co supervised in EPFL
- Courses in complex systems, deep learning, synthetic biology, quantum mechanics

MSc, Neuroscience — Bilkent University, Turkey / EPFL, Switzerland

- Courses in Computational Neuroscience, Deep Learning and NeuroAI

BSc, Molecular Biology and Genetics — Boğaziçi University, Turkey

- Courses on computer science, machine & deep learning, biomedical engineering, NLP

CERTIFICATIONS

Neo4j Certified Professional (2024), LLMOps & Fine-Tuning LLMs (DeepLearning.AI, 2024), Federated Fine-Tuning of LLMs with Private Data (DeepLearning.AI, 2025), AWS Certified Developer DVA-C02 (in progress), AWS Bedrock (2025), Quantization Fundamentals with Hugging Face (DeepLearning.AI, 2025), Artificial Intelligence on Microsoft Azure (2026), Infrastructure as Code in Google Cloud Platform (Linkedin Learning , 2024), Pyspark Essential Training: Introduction to Data Pipelines (2025), Practice it : Advanced SQL (Linkedin Learning, 2025), Github Copilot Technical & Advanced (2026), Machine Learning Professional (Databricks, in progress), Diffusion Tensor Imaging using DTI Studio and MRIcrogl (2017), Human Brain Project: Brain Simulation Platform (HBP, 2018)

TECHNICAL SKILLS

Languages & Frameworks: Python, Bash, SQL, Cypher, MATLAB; PyTorch, Numpy, JAX, TensorFlow, PyTorch Geometric, Hugging Face, Scikit-learn, Keras, Unsloth, PEFT, bitsandbytes, NEURON, NEST, SparkML, PySpark

GenAI & Agentic: MCP, A2A, LangChain, LangGraph, DSPy, CrewAI, FastAPI, FastMCP, MLflow 3, OpenRouter, Claude Code, Letta

AI Domains: NLP, Computer Vision, STT, TTS, Graph Machine Learning, Physically Inspired ML

Fine-Tuning & Optimization: PEFT, RLHF (DPO / PPO / GRPO), SFT, Quantization, ViT, ViT-LLM adapters, Hyperparameter Optimization: Optuna, Ray

Data & Retrieval: Neo4j, Weaviate, Milvus, Pinecone, Redis, PostgreSQL, MongoDB, Elasticsearch / OpenSearch, Databricks, Spark, Azure AI Search, Delta Tables

Cloud & Infrastructure: AWS, Azure (AI Foundry, Fabric, Unity Catalog), Docker, Terraform, GitHub / GitLab CI, Slurm / Dask HPC, Slurm, Databricks

Data Modalities: 3D mesh (obj, stl) and volumes (nrrd), sparse graph structures and knowledge graph standards (OWL, RDF, SHACL, Cypher) databases and platforms, connectomics & transcriptomics datasets, modeling for VR

LANGUAGES

English (C1) · French (B1) · Japanese (B1) · Turkish (Native)